VALENTINE GOLDEN GHANEM

Principal Biomedical Scientist | Medical Scientist | Epidemiologist

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PROFESSIONAL PROFILE

Medical laboratory scientist and public-health researcher whose work connects clinical diagnostic practice, laboratory quality, epidemiology and data science. Principal Biomedical Scientist at Cocoa Clinic, Ghana COCOBOD. Research interests include infectious-disease surveillance, spatial epidemiology, health inequities and the use of analytical methods to inform public-health practice.

PROFESSIONAL EXPERIENCE

Principal Biomedical Scientist — **Oct 2024 - Present**
Cocoa Clinic, Ghana COCOBOD Medical Department | Accra, Ghana
- Supervises clinical laboratory workflow during assigned shifts, including quality assurance, equipment management and result validation.

Senior Biomedical Scientist — **Jun 2020 - Oct 2024**
Cocoa Clinic, Ghana COCOBOD Medical Department | Accra, Ghana
- Supervised laboratory shifts and supported diagnostic validation, internal quality control, staff development and preparations for ISO 15189 accreditation.

Medical Laboratory Scientist, Community Medical Missions — **Jul 2015 - Present**
Ghana COCOBOD Medical Missions | Ghana
- Contributes to outreach planning and provides laboratory screening services during community medical programmes.

Volunteer Biomedical Scientist — **Jun 2014 - Jul 2015**
Life Medical Mission | Accra, Ghana
- Supported laboratory operations and diagnostic testing during community outreach clinics; contributed to standard operating procedures and quality systems.

Biomedical Scientist Intern — **Jul 2013 - Aug 2014**
Korle-Bu Teaching Hospital | Accra, Ghana
- Completed rotations in haematology, microbiology, chemical pathology and immunology, with exposure to SLIPTA and SLMTA quality-improvement frameworks.

EDUCATION

Master of Laws (LLM) in International Law - In progress — **2025 - Dec 2026**

Liverpool John Moores University, United Kingdom
Expected completion: December 2026

Master of Science in Public Health - Merit — **2023 - 2026**

University of Suffolk, United Kingdom
Focus: communicable-disease epidemiology, health-systems strengthening and public-health surveillance

Master of Science in Data Science - Distinction — **2023 - 2025**

University of East London, United Kingdom
Focus: machine-learning applications in public health

Bachelor of Science in Medical Laboratory Science — **2010 - 2013**

University of Ghana, Accra, Ghana
Microbiology specialisation; clinical rotations across principal diagnostic disciplines

RESEARCH, CREDENTIALS & METHODS

Selected scholarly work and professional record

SELECTED PUBLICATIONS AND RESEARCH OUTPUTS

2026 | PEER-REVIEWED
Spatial and Machine Learning Analysis of District-Level Health Insurance Inequities in Ghana
Cureus, 18(1), e101984. doi:10.7759/cureus.101984

2025 | PEER-REVIEWED
Socio-Behavioral and Spatial Determinants of HIV/AIDS Incidence in Ghana: An Ecological Cross-Sectional Study with Explainable Machine Learning
Advances in Infectious Diseases, 15(4), 669-694. doi:10.4236/aid.2025.154050

2025 | PEER-REVIEWED
HIV Prognosis in Sub-Saharan Africa: A Narrative Review of Mortality Trends, Structural Determinants, and Policy Implications
International Journal of Medical and Health Research, 3(2). doi:10.61424/ijmhr.v3i2.346

2026 | PREPRINT
Spatial Machine Learning and Longitudinal Analysis of Skilled Antenatal Care Access and Fertility-Related Inequities in Ghana (1988-2022)
medRxiv. Preprint; not peer reviewed. doi:10.64898/2026.07.16.26358217

2025 | PREPRINT - IN PEER REVIEW
Forecasting HIV/AIDS Incidence in Ghana: A Retrospective Observational Study Using Ensemble Machine Learning Models
Research Square. doi:10.21203/rs.3.rs-6639193/v1

AWARDS AND ACADEMIC RECOGNITION

The Industry of the Future Award | University of East London Alumni Achievement Awards, 2026. Named as a recipient in the University's official announcement.

Unicaf Africa Prize for Academic Excellence | Unicaf. Associated with public-health studies at the University of Suffolk.

PROFESSIONAL REGISTRATION AND MEMBERSHIP

Registration	Professional body
Medical Scientist | Ghana AHPC | 09007815	Medical Scientist | Ireland CORU | MS 063958
Epidemiologist A | Netherlands VvE | 1637	Fellow | Royal Society for Public Health | FRSPH 140437
Member | Academy of Clinical Science and Laboratory Medicine | 12474	Member | Ghana Association of Medical Laboratory Scientists | AS 1206

In progress: Certified in Public Health (CPH), National Board of Public Health Examiners - examination preparation in progress.

AREAS OF PRACTICE AND METHODS

Laboratory practice	Haematology, clinical chemistry, microbiology, immunology, diagnostic quality assurance and result validation.
Epidemiology	Infectious-disease surveillance, spatial epidemiology, health inequities and public-health research.
Analytical methods	Biostatistics, spatial autocorrelation, LISA, CART, Random Forest, XGBoost and SHAP.
Tools	Python, R, QGIS, SQL, Git, DHIMS2 and data visualisation.